

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

QUIZ-4

DURATION: 30 MINUTES

WINTER SEMESTER, 2022-2023

FULL MARKS: 15

Math 4543: Numerical Methods

Programmable calculators are not allowed. Write your Student ID at the top of your extra pages (if any).

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

STUDENT ID:

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- | | |
|--|-------------------------|
| 1. State the fundamental theorems of calculus. Applying these theorems, explain how Euler's method is used to approximate the value of a definite integral. | 3 + 2
(CO4)
(PO1) |
| 2. Compare and contrast the following numerical algorithms for solving 1st order ODEs – Euler's method, Runge-Kutta 2nd order method, and Runge-Kutta 4th order method. | 2
(CO3)
(PO1) |
| 3. Determine the approximate value of
$I = \int_8^{30} \left[2000 \ln \left(\frac{140000}{140000 - 2100t} \right) - 9.8t \right] dt$ using Simpson's 1/3 rule (with $n_1 = 6$) and Simpson's 3/8 rule (with $n_2 = 6$). | 6
(CO2)
(PO1) |
| 4. An internet meme is defined as a humorous or culturally relevant image, video, or text that spreads rapidly online, often evolving as it is shared and adapted by users. Draw a rough sketch of a math meme related to this course. | 2
(CO69)
(PO420) |
- Warning:** Cringe and lame memes will result in negative marking.